in Joshimath. For two years now, locals had been warning the administration about the growing incidents of land subsidence and houses developing cracks. Some 14 months ago, they even formed a 'Joshimath Bachao Sangharsh Samiti'. The state government constituted an expert panel to conduct geological and geotechnical investigations, but even its wise cautionary notes went in vain.

The last week of December 2022 saw their nightmare come true as houses, hotels and even roads developed huge cracks and water started seeping up through some of them—almost like the Himalayan town had turned into the RMS *Titanic*. Chief minister Pushkar Singh Dhami visited the town on January 7 and has since announced an interim relief of Rs 1.5 lakh to each affected family. In a town of just over 20,000, some 723 houses have been declared unsafe for habitation so far. Kicking off a phased process, demolition work will start on the 75 most vulnerable buildings under the supervision of scientists from the Central Building Research Institute (CBRI), Roorkee. The demolition work did not start on January 10 due to protests by locals. The residents, including hoteliers, say they were not served any notice and are demanding a proper survey of their damaged properties.

Politicians of all hues have made a beeline for Joshimath, but what residents are angry about is the over-ambitious 'development' policy they had rolled out earlier. Specifically, the locals are demanding the closure of two ongoing mega projects in the town's vicinity: a tunnel being blasted through the hills for the 520 MW Tapovan-Vishnugad hydel project and the two-lane Helang-Marwadi Bypass being built as part of the Char Dham highway to Badrinath. "The unscientific approach in pursuing mega construction projects has led us to this situation. We need immediate remedial steps and long-term planning," says Atul Sati, Samiti convenor.

History should have been warning enough. In 1976, in a continuing policy rethink following the devastating floods of 1970, the Uttar Pradesh government had set up an 18-member committee headed by then Garhwal commissioner M.C. Mishra. The committee ruled that Joshimath is situated on the settled moraine of a landslide, with the tectonic bedrock located much deeper inside, and as such is prone to landslides. Not to speak of the seismic threat—that rock is part of what's called the Main Central Thrust, the point where the Indian landmass continues to push into the Eurasian plate, and is hence by definition unstable. The committee urged no interference with the hill slopes, that trees not be cut, and a proper drainage be built for the town. None of the three recommendations has been followed.

IT'S NOT JUST JOSHIMATH THAT FACES RUIN. MANY HILL STATIONS LIKE NAINITAL, MUSSOORIE, TOO, FACE SUBSIDENCE ISSUES

Over the years, the area saw an unbridled construction boom to cater to the rush of tourists and pilgrims. Over 1.5 million pilgrims visited Badrinath in just 2022. The expert panel last year—with specialists from the Uttarakhand State Disaster Management Authority (USDMA), CBRI Roorkee, IIT Roorkee, Wadia Institute (Dehradun) and the Geological Survey of India—also reiterated that the town was built on an unstable foundation that can give way in case of heavy rains, tremors, unregulated construction or footfalls far beyond the town's carrying capacity. It attributed the subsidence to the buildings that have mushroomed on the Joshimath-Auli route, aggravated by poor drainage and riverbank erosion.

A central team of scientists is now in Joshimath to carry out a survey. A complete ban has been imposed on construction activity and affected residents are being relocated to safer buildings—over 4,000 people have had to leave their homes in the biting cold, and now face an uncertain future. As temporary relief, the state has ordered that they be provided with meals, besides a monthly dole of Rs 4,000. A National Disaster Management Authority (NDMA) team has arrived in Dehradun to advise the CM, and four companies of the National Disaster Response Force (NDRF) have also reached Joshimath.

This momentary rush of crisis management won't solve the larger scourge, though. It is not only Joshimath that faces possible ruin. Many areas in the famous hill stations of Mussoorie and Nainital are facing subsidence; many more like Karnaprayag, numerous villages in Rudraprayag district, many parts of Uttarkashi district that faced landslides from the Varunavrat mountains over a decade ago, are staring at a similar fate. Blame unbridled, unplanned 'development' in zones that are already ecologically and geologically fragile. CM Dhami has now ordered a technical survey of all hill towns to assess their bearing capacity; construction will be banned if it crosses limits.

A region that has witnessed frequent natural disasters like earthquakes, landslides, cloudbursts and flash floods saw its worst deluge at Kedarnath in 2013, where over 5,000 pilgrims and tourists lost their lives while infrastructure throughout the state suffered unprecedented destruction. In 2021, over 300 people lost their lives in natural disasters. Dharchula in Pithoragarh district was hit by huge landslides from the Aildhara hills overlooking the town. Residents there also live in fear of flash floods if the protection wall along the Kali river, which forms the border between India and Nepal, is not constructed before the monsoon. The state had identified 465 villages vulnerable to natural calamities that needed relocation by 2021; it has managed only 1,100 families in 44 villages till now. ■